

1.0 Operation Procedure

Step 1 – Turn on switch

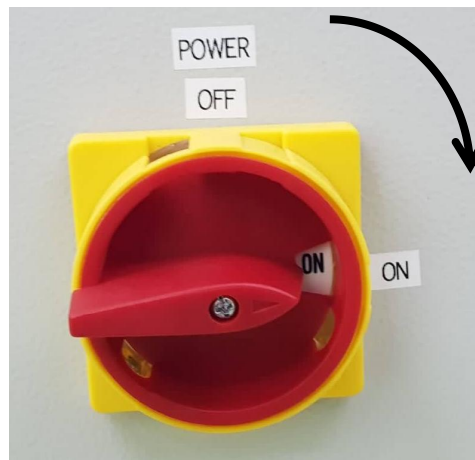


Figure 1.0.1 On/Off Switch

Step 2 - Click Reset Button when the system is started and alarm buzzing

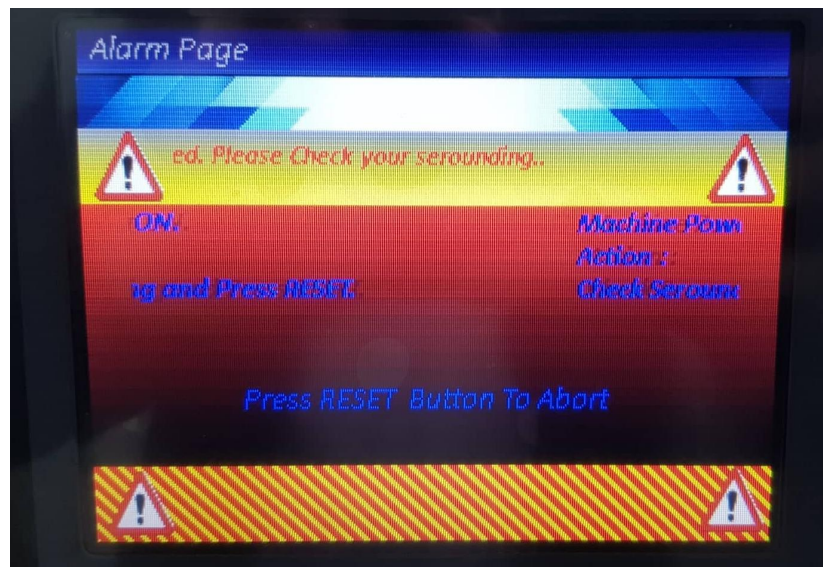


Figure 1.0.2 Alarm Page when system starts



Figure 1.0.3 Physical Buttons – Start Button, Reset Button and Emergency Button

Step 3 – Confirm all the configuration settings, holders in place and click the ready button in the screen



Figure 1.0.4 Auto Mode Interface



Figure 1.0.5 System Ready

Step 4 – When the ready button turns green, click start button to start the operation and wait for the process to end, then remove the holder from the machine. If the window is open during the process, the alarm will be triggered and user can choose either continue the process by clicking the start button or cancel the operation by clicking the reset button

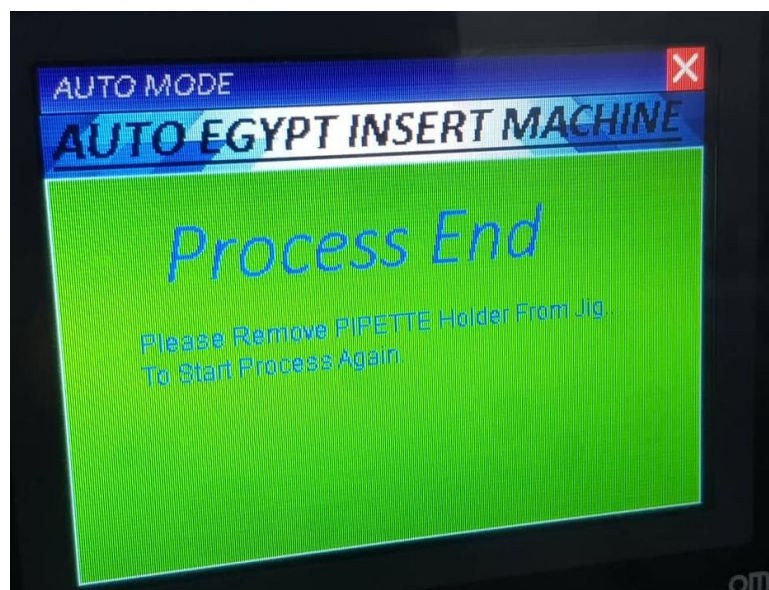


Figure 1.0.6 Process End Interface

2.0 System Interface

2.1 Main Menu



Figure 2.1.1 Main Menu

Items	Description
Auto Page	Go to Auto Mode Page
Maintenance Page	Go to Maintenance Menu Page
Setting Page	Go to Setting Menu Page
Alarm History	Go to Alarm History

2.2 Auto Mode



Figure 2.1.1 Auto Mode Interface

Description:

This interface will be displayed when the system is started up. In this interface, the progress and position of the operation is displayed.

Items	Description
Ready	Confirm system is ready for operation
Maintenance Mode	Go to Maintenance Setting
Selection Mode	Go to Selection Mode
Alarm History	Go to Alarm History
MC Speed	Modify operation speed
Cycle Time	Operation taken time
Process Count	Accumulated number of process
Reset Count	Reset accumulated number of process
Close / X	Exit to Main Menu

2.3.0 Maintenance Mode

2.3.1 Maintenance Menu

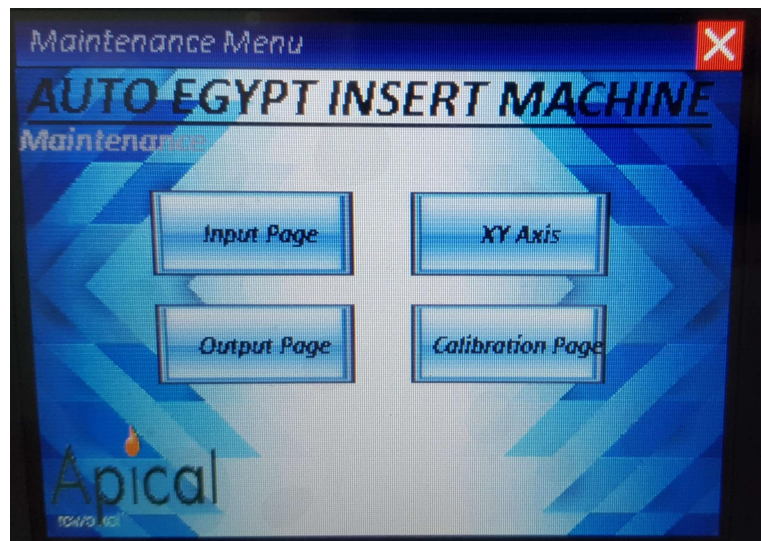


Figure 2.2.1 Maintenance Menu

Items	Description
Input Page	Go to input settings page
Output Page	Go to output settings page
XY Axis	Go to XY Axis settings page
Close / X	Exit to Main Menu

2.3.2 Input Page



Figure 2.2.2.1 Input Page 1



Figure 2.2.2.2 Input Page 2

Description:

This Input Page interface is used to check the input for the system.

Items	Description
Next	Go to next page
Back	Go to previous page
Close / X	Exit to Main Menu

2.3.3 Output Page



Figure 2.2.3.1 Output Page 1



Figure 2.2.3.2 Output Page 2

Description:

This Output Page interface is used to check the output of the system is working properly.

Items	Description
Next	Go to next page
Back	Go to previous page
On	Turn off the setting
Off	Turn on the setting
Close / X	Exit to Main Menu

2.3.4 XY Axis



Figure 2.2.4.1 XY Axis Menu Page 1

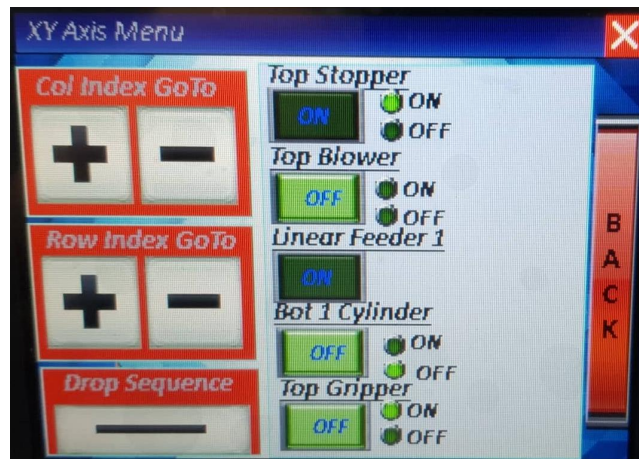


Figure 2.2.4.2 XY Axis Menu Page 2

Description:

This XY Axis interface is used to configure the XY axis and position for the platform in the machine.

Items	Description
Next (Page 1)	Go to next page
Back (Page 2)	Go to previous page
Up / Down / Left / Right (Page 1)	Modify X and Y axis
GoTo (Page 1)	Instruct the platform to go to the specific position
Home (Page 1)	Instruct the platform to go to the default position
Save (Page 1)	Save the XY Axis configuration
Plus (+) (Page 2)	Increment Row / Column Index
Minus (-) (Page 2)	Decrement Row / Column Index
Drop Sequence (-----)(Page 2)	Drop the current sequence and return to initial sequence
On (Page 2)	Turn off the setting
Off (Page 2)	Turn on the setting
Close / X	Exit to Main Menu

2.4.0 Setting Page

2.4.1 Setting Menu



Figure 2.3.1.1 Setting Menu

Items	Description
Offset Setting	Go to Offset setting
XY Axis Setting	Go to XY Axis setting
Section Setting	Go to Section setting

2.4.2 Offset Setting



Figure 2.3.2.1 Offset Setting

Description:

This offset setting interface is used to configure the offset between rows and columns to make sure the egypt can be placed properly.

Items	Description
Row offset	Configure the row offset
Column offset	Configure the column offset

2.4.3 XY Axis Setting

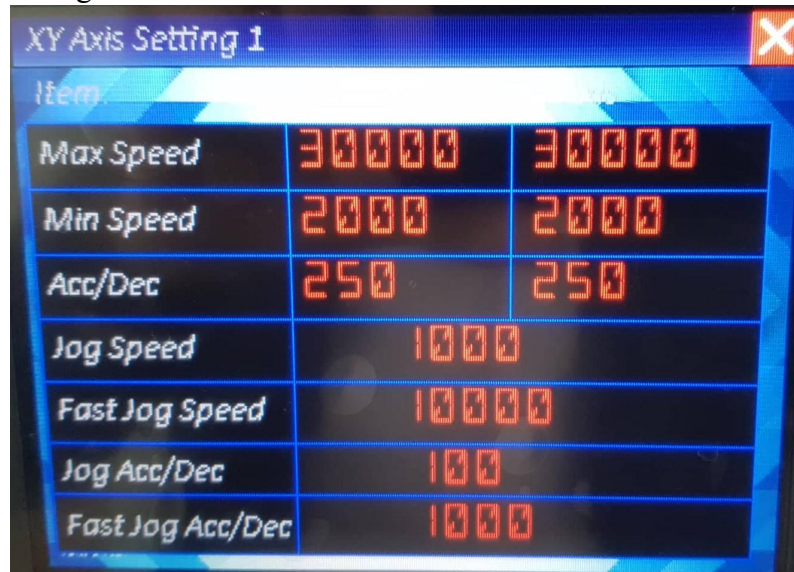


Figure 2.3.3.1 XY Axis Setting

Description:

This XY Axis Setting interface is used to configure the operation speed of each hardware in the system.

2.4.4 Section Setting



Figure 2.3.4.1 Section Setting

This Section Setting interface is used to configure the total number of sections available and the total number of column and row on the holders.

2.5.0 Alarm History

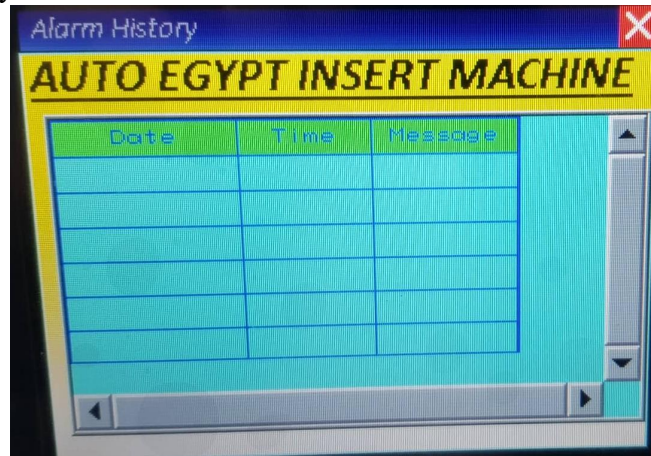


Figure 2.4.0.1 Alarm History

Description:

This Alarm History interface display the log when the alarm is triggered and the specific message.

2.6.0 Selection Mode



Figure 2.5.0 Selection Mode

Description:

This Selection Mode interface is used to select a particular section to be inserted, all the sections are enabled by default, click on the section to disable it and click on it again to enable it back.

2.7.0 Alarm Page

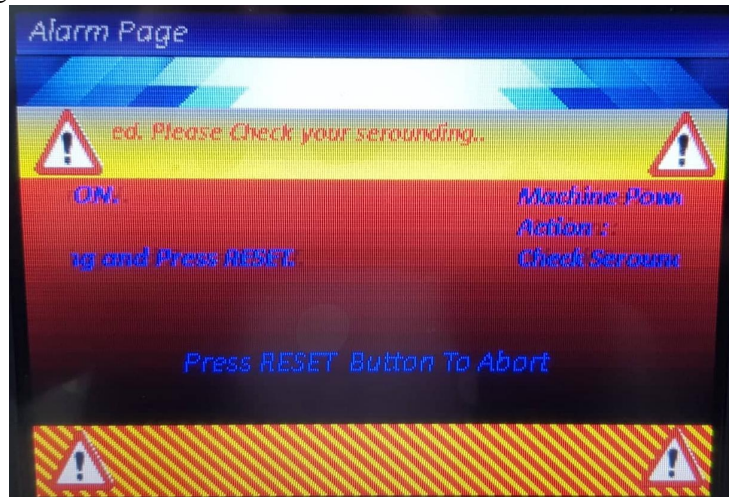


Figure 2.7.0.1 Alarm Page

Description:

Alarm page will appear when the system initiating and user is required to click the physical reset button to close it. Then, the alarm page will appear during the insert operation if the window is open manually, the user can choose to continue the operation by closing the window and click the physical start button or reset the operation by clicking the physical reset button.

2.8.0 Process End Page



Figure 2.8.0.1 Process End Page

Description:

The process end page will appear after the operation end and user is prompt to remove the holder from the machine.

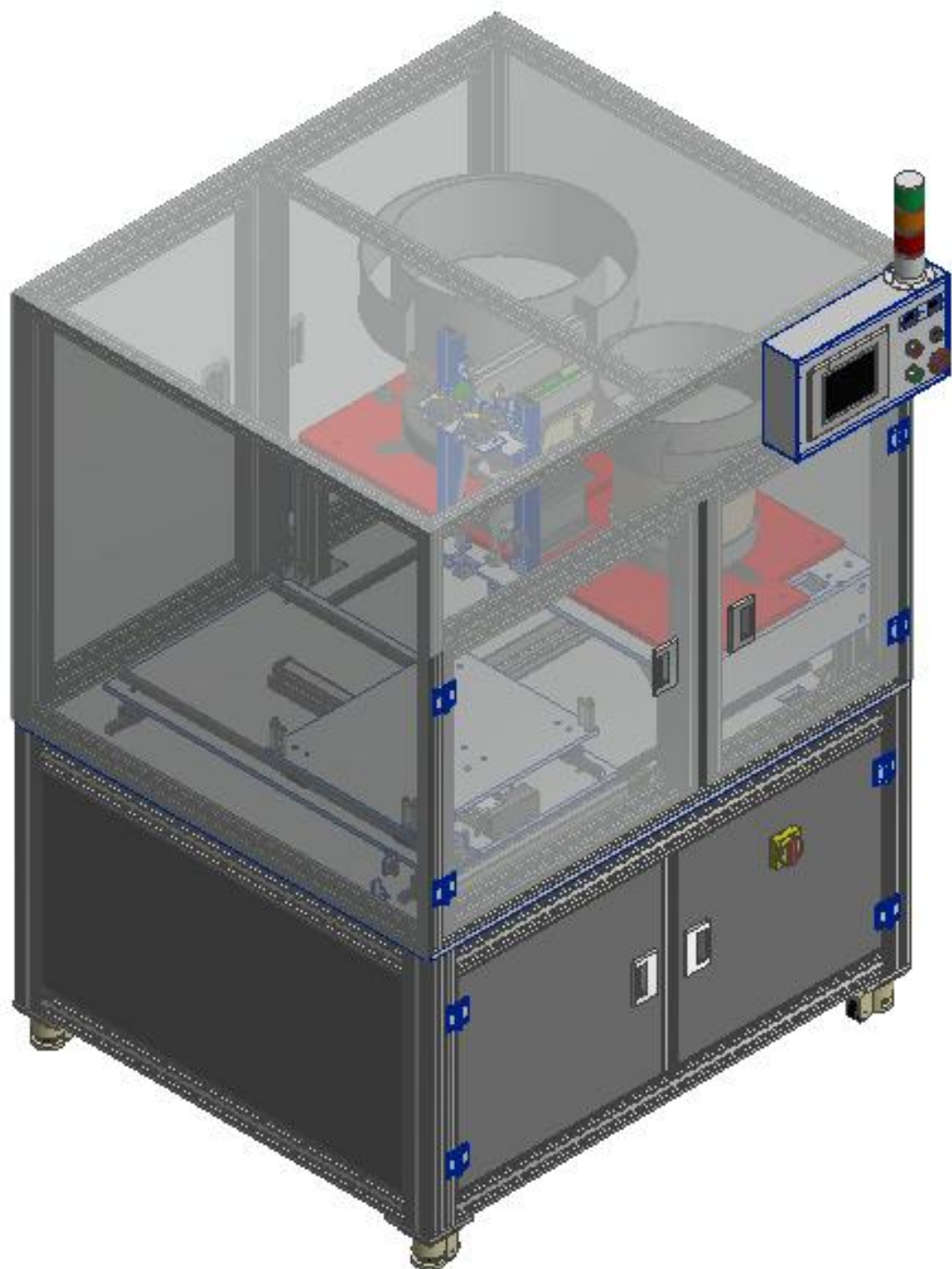
MACHINE INTRODUCTION

This machine is an Automatic Pippete Tip Insertion Machine.

Made to reduce the tact time of using manual labor insertion.

Bowl feeder are used to sorted the pippete tips before insertion to holder case.

Pneumatic actuators and Electric actuators are used for the insertion process.

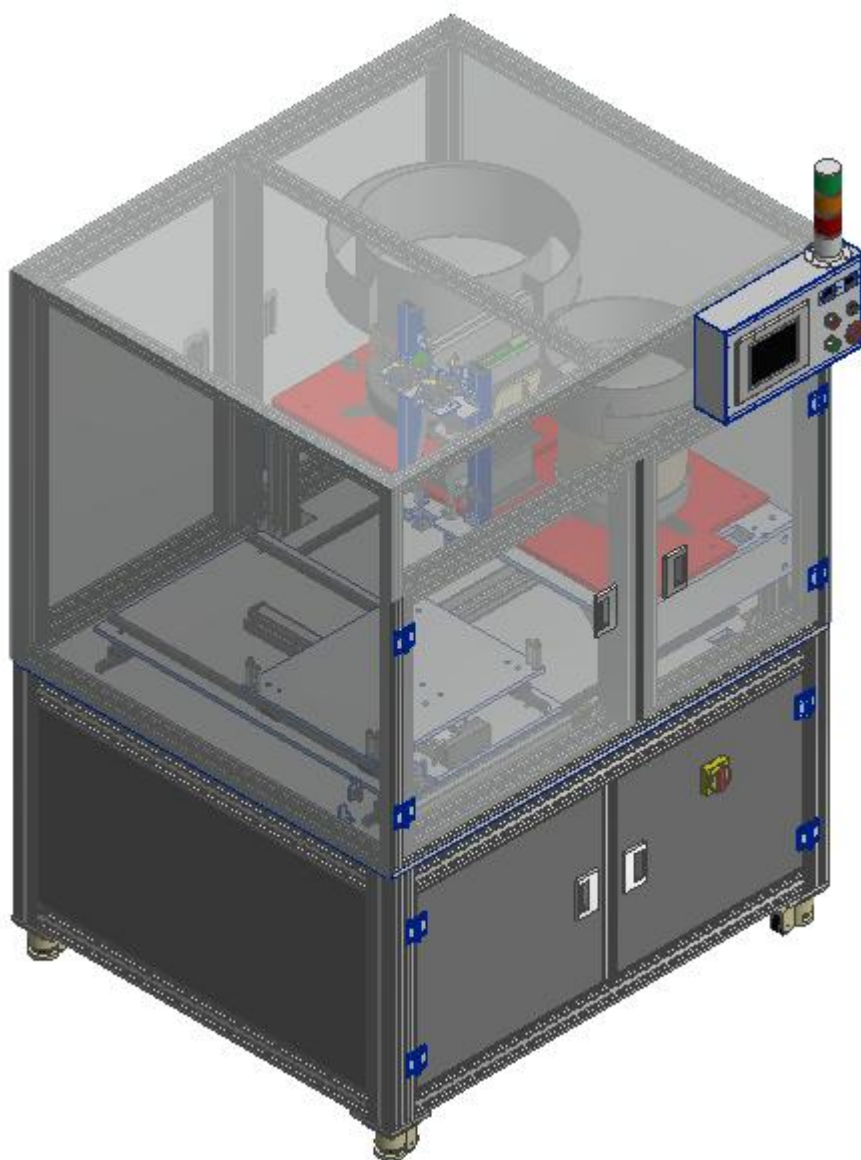




YOSHIDEN

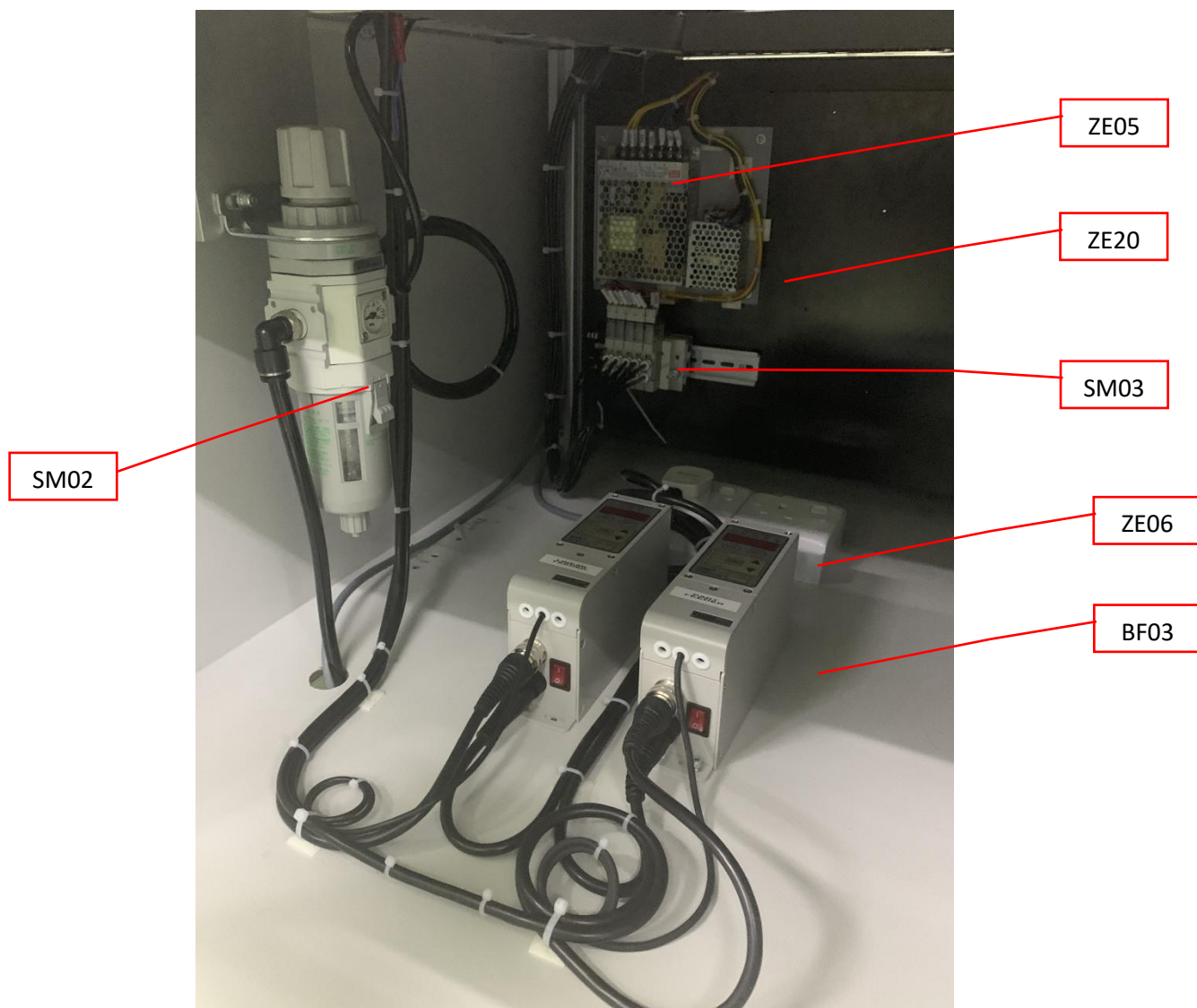
AUTOMATIC

PIPPETE TIPS INSERTION MACHINE

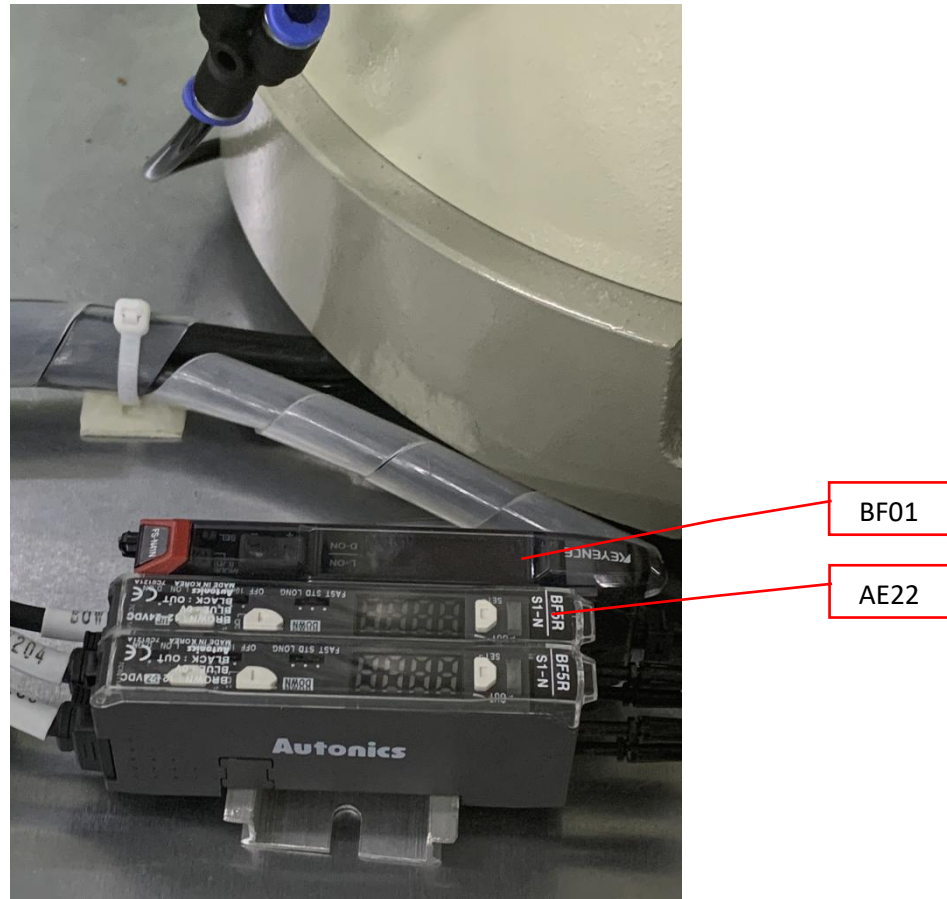


USER MANUAL

MECHANICAL PART LIST



ITEM No.	DESCRIPTION
SM02	AIR REGULATOR
SM03	SOLENOID VALVE
BF03	FEEDER AMPLIFIER
ZE05	POWER SUPPLY
ZE06	3 PIN SOCKET
ZE20	POWER SUPPLY



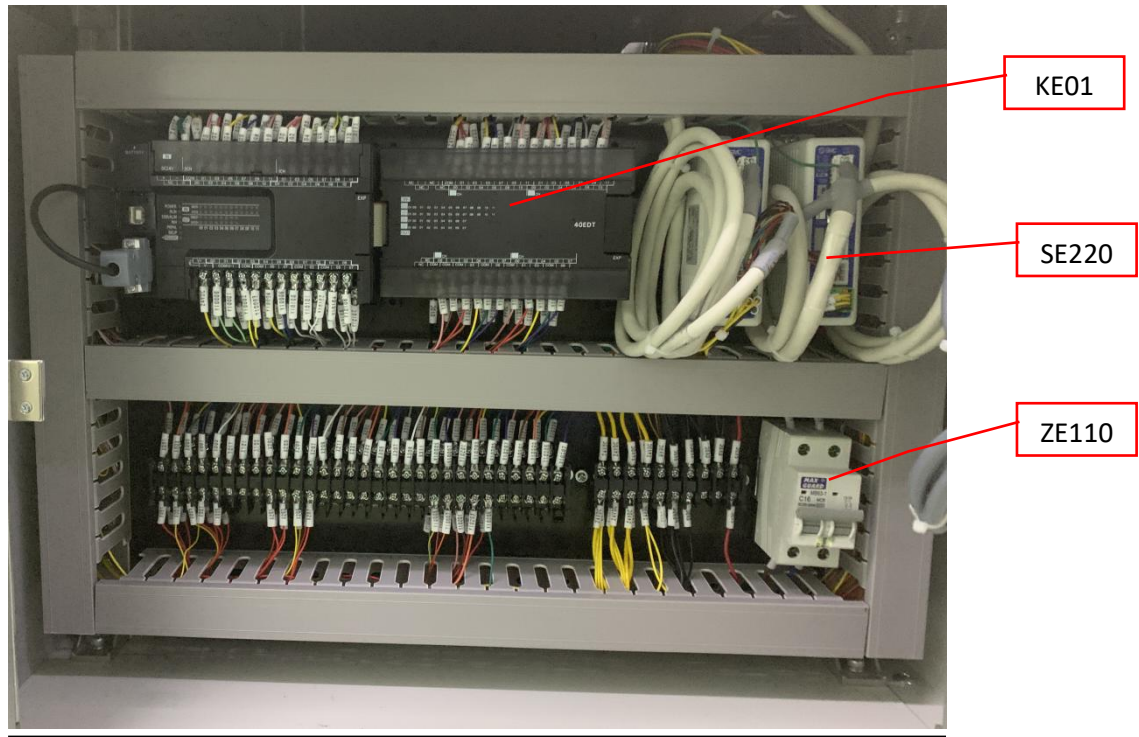
ITEM No.	DESCRIPTION
BF01	SENSOR AMPLIFIER
AE02	SENSOR AMPLIFIER

OPERATION MANUAL



ITEM No.	DESCRIPTION
FA01	AIR COMPRESSOR

ELECTRICAL & PLC WIRING LIST



ITEM NO	DESCRIPTION
KE01	PLC
SE220	ACTUATOR DRIVER
ZE110	CIRCUIT BREAKER